TrES-5b

R-0839

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_TRES-5_180415` · created 2026-07-28T18:52:20+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458223.9434 +/- 0.0026 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.119 +/- 0.037
Transit depth (Rp/Rs)^2	1.41 +/- 0.88 [%]
Orbital Inclination (inc)	82.64 +/- 2.44
Airmass coefficient 1 (a1)	0.945 +/- 0.014
Airmass coefficient 2 (a2)	0.0341 +/- 0.009
Scatter in the residuals of the lightcurve fit is	1.51 %
Best Comparison Star	#3 - [60, 294]
Optimal Method	PSF photometry
Transit Duration (day)	0.054 +/- 0.025

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.119 ± 0.037 vs 0.142 (deviation 0.62σ)
Duration fit vs published	0.054 d vs 0.0758 d (deviation 0.86σ)
Mid-time O–C	-6.5 min
Depth SNR	3.2
Residual scatter	1.51 %

Light curve

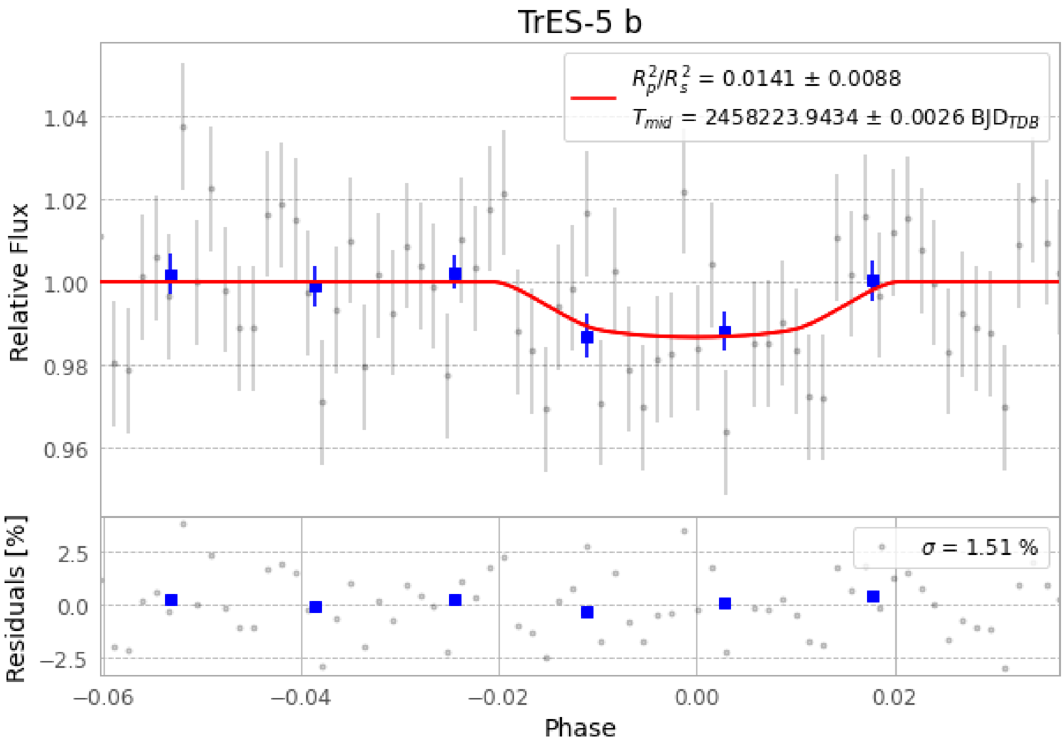

Figure 1 — FinalLightCurve_TrES-5 b_2018-04-15.png (SHA-256 2bc4b636029136c5…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [462, 210], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 6fe47e57ed66d61a…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9`

Data provenance

70 FITS frames (46 MB), `TRES-5180415083012.FITS` → `TRES-5180415115712.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-180415.log (1799d26558b3…), FinalLightCurve_TrES-5 b_2018-04-15.png (2bc4b6360291…), FinalLightCurve_TrES-5 b_2018-04-15.csv (1129d4ac37ee…)

Compute resources

Wall time 115.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)